

Modeling Pragmatic Reasoning behind Social Meaning

[Anonymous for review]

Abstract

Researchers are increasingly interested in how pragmatic reasoning influences the social impressions we form of a speaker. This study examines the complex interplay between language use, situational context, pragmatic inference, and social meaning. The Social Evaluation Model (SEM) is presented, a probabilistic framework that models how listeners form social judgments about speakers based on their linguistic choices and the situational context, by reasoning about the speaker’s knowledge states and communicative motivations. We situate SEM within the broader tradition of probabilistic pragmatics, showing how it extends the inferential chain of the Rational Speech Act framework to social evaluation as an explicit target of inference. We test the model’s predictions using data from two experiments on numerical (im)precision and demonstrate its application to pragmatic norm violations. Our findings contribute to existing research in sociolinguistics and formal pragmatics, providing a principled probabilistic framework for understanding how linguistic choices convey social meaning through pragmatic inference.

Keywords: social meaning, pragmatic inference, social evaluation, probabilistic pragmatics, Rational Speech Act, numerical precision

1 Introduction

The study of pragmatic inference—how listeners derive meaning beyond the literal words—began with H. P. Grice’s influential work on implicature (Grice, 1975). Grice proposed

that conversation follows a *Cooperative Principle* along with maxims that speakers typically adhere to for effective communication. When a speaker seemingly violates a maxim, the listener assumes cooperation and infers an additional layer of meaning, known as *implicature*. This framework established that meaning relies on *rational inference* regarding the speaker’s intent. Building on Grice’s foundation, Horn (1989) and Levinson (2000) introduced *neo-Gricean* models, refining the types of implicature and incorporating scalar reasoning (e.g., “some” suggesting “not all”). These early theories demonstrated that pragmatic inference involves not just language but also social reasoning, shaped by expectations of cooperation and relevance.

Gricean pragmatics initially focused on cooperative reasoning, but later developments began to view implicature as indicative of the speaker’s epistemic state—what the speaker knows, believes, or is committed to. Sauerland (2004) and Spector (2007) differentiated between strong and weak scalar implicatures, demonstrating that the listener’s inference hinges on whether the speaker knows a stronger alternative to be false or simply does not know it to be true. This represented a shift from truth-conditional to epistemic interpretations of pragmatic reasoning. Experimental and formal studies, such as those by Chemla and Singh (2014), supported the notion that many implicatures rely on evaluating the speaker’s knowledge state. Probabilistic and Bayesian models, including the Rational Speech Act (RSA) framework (Frank and Goodman, 2012; Goodman and Stuhlmüller, 2013; Potts et al., 2016), advanced this perspective by conceptualizing pragmatic inference as Bayesian reasoning over the speaker’s epistemic beliefs and communicative goals. From this viewpoint, pragmatic inference is nuanced and probabilistic, reflecting uncertainty about the speaker’s knowledge and intentions.

Beyond epistemic reasoning, a significant area of research highlights that pragmatic inference involves uncovering speaker intentions and motives—the *why* behind an utterance. Grice (1957) defined meaning through intention recognition. Sperber and Wilson’s (1995) Relevance Theory portrays communication as the process of inferring a speaker’s communicative and informative intentions. Sociolinguistic and cognitive perspectives (e.g., Brown and Levinson, 1987; Tomasello, 2003) emphasize the influence of social motives—

such as politeness, persuasion, and face-saving—on pragmatic inference. Empirical studies on indirect speech acts (e.g., Gibbs, 1986; Lee and Pinker, 2010) show that listeners discern not only what speakers mean but why they opt for indirect or strategic expressions, revealing motives like maintaining deniability or managing social relationships.

Recent computational and cognitive models unify these concerns, viewing pragmatic inference as joint reasoning about epistemic states and communicative intentions. The RSA framework (Frank and Goodman, 2012; Goodman and Frank, 2016) explicitly combines these dimensions: listeners assess the likelihood of an utterance given the speaker’s knowledge, goals, and motivations. Bergen, Levy, and Goodman (2016) conceptualize pragmatic understanding as probabilistic reasoning involving both semantic alternatives and speaker motives, linking epistemic and intentional inference. This convergence illustrates a wider trend in contemporary pragmatics, which integrates informational (epistemic) and intentional (motivational) elements of meaning as parts of a unified social-cognitive process.

Research increasingly shows that contextual factors—such as prior discourse, shared knowledge, expectations, and social settings—significantly influence pragmatic inferences. Studies in experimental pragmatics (e.g., Breheny, Katsos, and Williams, 2006; Degen and Tanenhaus, 2015) reveal that the availability and prominence of alternatives depends heavily on contextual relevance and mutual assumptions between speakers. Rooij (2003) and Clark (1996) stress that pragmatic reasoning occurs within common ground and joint activity, indicating that the same utterance can lead to different inferences depending on the conversational or cultural context. This context sensitivity illustrates that pragmatic inference is not a rigid mechanism but a flexible reasoning process attuned to social, informational, and situational cues.

Recent research in sociopragmatics explores how pragmatic reasoning interacts with *social meaning*—the information language provides about a speaker’s social identity, affiliations, and personal traits (Eckert, 2008; Podesva, 2011; Silverstein, 2003). The foundation of social meaning research stems from Labov’s (1972) demonstration that linguistic variation is systematic and socially meaningful, linking language use to social structure

and change. Building on this, Eckert’s (2012) “Three Waves” framework illustrates the field’s evolution from correlating variables with demographic categories to understanding stylistic practice, stance, and persona as sites where social meaning is constructed.

This sociolinguistic foundation has recently aligned with pragmatic and semantic perspectives. Research on sociophonetic variation (Burnett, 2017; Burnett, 2019; Burnett, 2023; Burnett, 2025), the pragmatics of the definite article (Acton, 2019), and inferences about relevance and informativity (Beltrama and Papafragou, 2023) demonstrates that context, identity, and pragmatic reasoning collaboratively shape meaning. Collectively, this research underscores the intricate connection between pragmatic inference and social interpretation: listeners consider not only *what* speakers mean and *why* they say it, but also *who they are* and *how they position themselves* in conversation.

In summary, the development from Grice to modern computational and sociopragmatic approaches reveals a gradual expansion of what pragmatic inference involves—from cooperative logic, to reasoning about epistemic states, to modeling intentions, motives, and social meanings. Pragmatic understanding is a form of social cognition shaped by contextual constraints: listeners must reason about what the speaker knows, what they intend to achieve, how their utterance affects their social position, and how the surrounding context affects these interpretations.

Building on these insights, we introduce the Social Evaluation Model (SEM), a probabilistic framework that formalizes how linguistic choices, situational context, and pragmatic reasoning work together to shape listeners’ social judgments. SEM shows how listeners infer a speaker’s knowledge state and communicative motivation from their language use, and how these inferences give rise to evaluations of social attributes such as competence, reliability, and likability. The framework expands the inferential chain of probabilistic pragmatics to include evaluative meaning, where small differences in language use can lead to subtle, context-dependent changes in social interpretation (e.g., Beltrama and Papafragou, 2023).

The paper is organized as follows. Section 2 provides a formal definition of the SEM, situating it within the probabilistic pragmatics tradition and comparing it to the RSA

framework. A case study on numerical (im)precision is used to develop and qualitatively evaluate the model in Section 3. Section 4 presents a quantitative evaluation of the model’s predictions, including a comparison of the full model against nested ablations. Section 5 discusses the model’s contributions and directions for future work.

2 The Social Evaluation Model

The Social Evaluation Model (SEM) is a probabilistic framework for predicting how listeners form social judgments about speakers based on their linguistic choices. It builds on the core architecture of probabilistic pragmatics, in particular the Rational Speech Act framework (Frank and Goodman, 2012; Goodman and Frank, 2016) and the broader tradition of Bayesian pragmatic modeling surveyed by Franke and Jäger (2016), while extending that architecture toward a target of inference that has not previously been formalized within it: the social evaluation of the speaker as a communicative agent.

Like RSA and related models, SEM treats pragmatic interpretation as Bayesian inference. A listener who hears utterance v in context c does not simply decode its semantic content; they reason probabilistically about what mental states could have produced that utterance. Central to this reasoning is a set of utterance alternatives V —the other things the speaker could have said—and a commitment to the truthfulness constraint, which limits the knowledge states a speaker could plausibly be in given what they actually said. These features are not incidental borrowings from RSA; they are load-bearing components of SEM, and the model’s predictions depend on them in the same structural way RSA predictions do.

Where SEM departs from existing probabilistic pragmatic models is in what the listener is ultimately trying to infer. In RSA, inference terminates at meaning: the listener recovers the speaker’s intended message. In SEM, meaning recovery is not the endpoint but an intermediate step. The listener uses their inferences about the speaker’s epistemic state—what the speaker knew, and what they could therefore truthfully have said—to make a further inference about the speaker’s communicative motivation: why did they choose this utterance over its alternatives? And from that motivational inference, the

listener arrives at a social evaluation: what does this choice reveal about the speaker’s competence, reliability, or warmth?

This staged reasoning—from utterance to knowledge state, from knowledge state to motivation, from motivation to social attribute—is what SEM formalizes. The key novel components, relative to RSA, are: (i) social attributes as an explicit target of inference; (ii) discrete motivation-based strategies, each defined as a function from context and knowledge state to utterance, which are more enumerable and empirically grounded than the utility functions in RSA; (iii) an impact function $I(a|k, m; \omega)$ that links epistemic and motivational inferences to graded social evaluations; and (iv) empirically fitted trivalent functions that specify the direction of each inference-to-evaluation link. There is no counterpart to this function in standard RSA, because RSA does not model social evaluation as an output. Figure 1 provides an overview of the shared and novel components, and Figure 2 shows the full model as a Bayesian network.

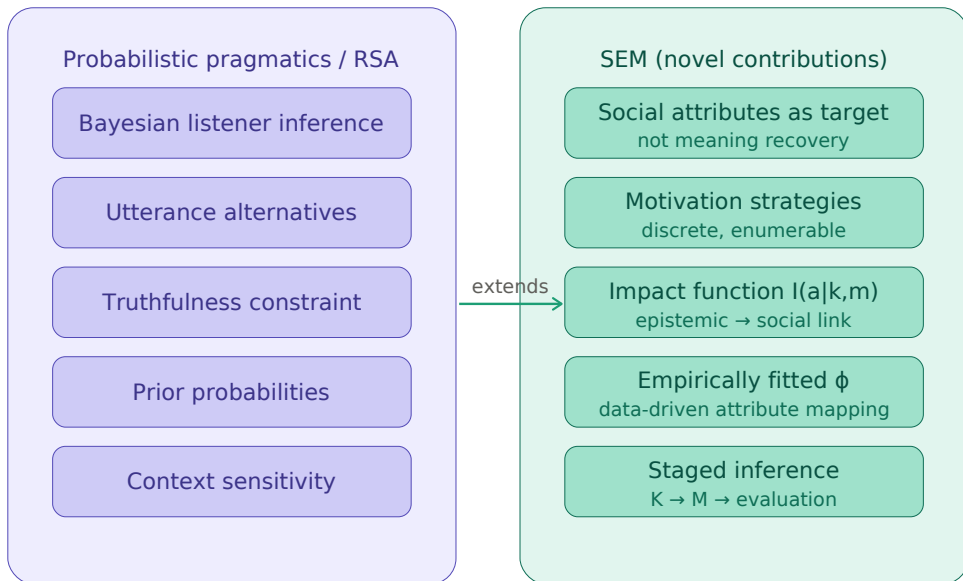


Figure 1: The relationship between Probabilistic Pragmatics / RSA and the Social Evaluation Model (SEM). SEM extends the core components of RSA—Bayesian listener inference, utterance alternatives, truthfulness constraint, prior probabilities, and context sensitivity—with novel components: social attributes as the target of inference, discrete motivation strategies, an impact function $I(a | k, m; \omega)$ linking epistemic to social dimensions, empirically fitted trivalent functions ϕ , and staged inference ($v, c \rightarrow K \rightarrow M \rightarrow a$).

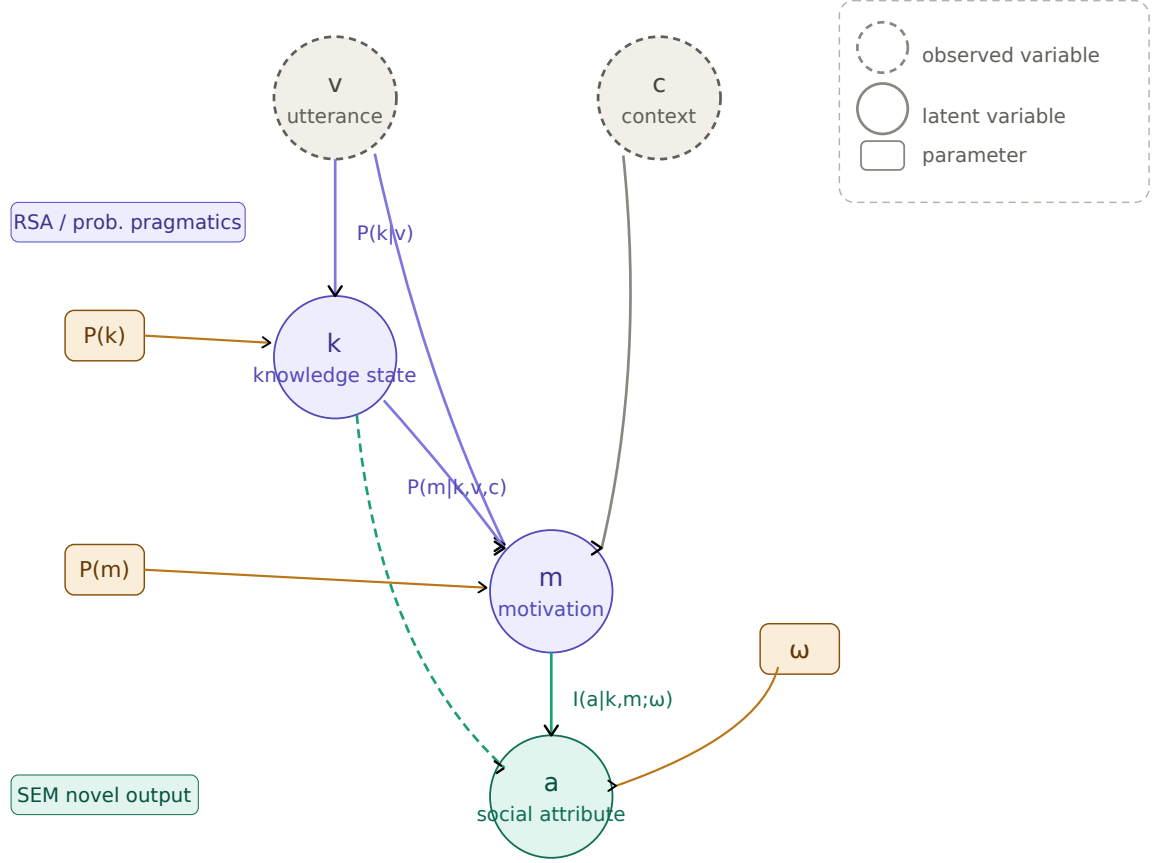


Figure 2: Bayesian network representation of the Social Evaluation Model. Dashed circles denote observed variables (utterance v , context c) available to the listener. Solid circles denote latent variables inferred by the listener: knowledge state k , motivation-based strategy m , and social attribute evaluation a . Rectangles denote fixed parameters: prior probabilities $P(k)$ and $P(m)$, and the weight parameter ω . Inference proceeds in three stages: v and $P(k)$ jointly determine the posterior over knowledge states $P(k|v)$; k , v , c , and $P(m)$ determine the posterior over strategies $P(m|k, v, c)$; and k and m feed into the impact function, weighted by ω , to yield the social evaluation a . The dashed arrow from k to a reflects the direct contribution of epistemic inference alongside the motivational pathway via m .

2.1 The Model Architecture

The social evaluation model¹ (SEM) predicts the listener’s social assessment of the speaker based on using utterance v from a set of alternatives V , given a situational context $c \in C$. Formally, the listener’s evaluation function $E_L(a|v, c)$,² which specifies how the listener evaluates a speaker concerning social attribute a , is defined as:

¹A Python implementation of SEM and its applications to the case studies can be found here: [LINK TO ADD].

²The function is defined as $E_L : A \times V \times C \rightarrow \mathbb{R}$; it returns a numerical score representing the listener’s evaluation of the speaker on attribute a .

$$E_L(a|v, c) = \sum_{k \in K} \sum_{m \in M} P(k|v) \cdot P(m|k, v, c) \cdot I(a|k, m; \omega) \quad (1)$$

where K is a set of speaker knowledge states, M is a set of motivation-based strategies,³ $P(k|v)$ and $P(m|k, v, c)$ are conditional probabilities, and $I(a|k, m; \omega)$ is an impact function that determines how inferences about knowledge state and motivation jointly affect the listener’s evaluation of social attribute a . The weight factor ω adjusts the relative contribution of epistemic versus motivational inference to the evaluation.

This equation can be understood as formalizing two intuitive notions: (i) when a listener hears utterance v in context c , they infer the probability of the speaker being in knowledge state k and the probability of the speaker acting according to motivation strategy m ; and (ii) the listener holds an internal mapping from knowledge states and motivations to social attributes, captured by the impact function I . The evaluation function E_L returns a real number $x \in [-1, 1]$, where $x = -1$ indicates the lowest evaluation and $x = 1$ the highest.⁴

Knowledge states and the truthfulness constraint. A central aspect of SEM is the set V , which contains the utterances a speaker considers as alternatives in a given situation. SEM focuses specifically on *non-equivalent* alternatives, i.e., forms that differ in their core-semantic meaning. This is important because semantic differences between alternatives create distinct constraints on what knowledge state the speaker could plausibly be in: a speaker who uses a semantically weaker form (e.g., “some students passed”) may or may not know the stronger fact, whereas a speaker who uses a semantically stronger form (e.g., “some but not all students passed”) is committed to knowing it.

These different epistemic possibilities are represented by the set of knowledge states K . Inferences about knowledge states are formalized by the conditional probability $P(k|v)$,

³A motivation-based strategy is a function that specifies what utterance a speaker would produce across different contexts and knowledge states when driven by a particular motivation. The same behavioral strategy can in principle arise from different underlying motivations; the case study in Section 3 provides detailed examples.

⁴When a categorical judgment is required (e.g., does the speaker possess attribute a or not), one applies the sign function: $x_{cat} = \text{sgn}(x)$. For a probabilistic output, one applies the sigmoid: $x_{prob} = (1 + e^{-\kappa x})^{-1}$, where $\kappa > 0$ controls the steepness.

derived from the assumption that the speaker adheres to the maxim of truthfulness: they only assert what they know to be true. This assumption is implemented via a truthfulness function $t : K \rightarrow \mathcal{P}(V)$, which maps each knowledge state k to the set of utterances the speaker could truthfully produce in that state. The posterior probability of knowledge state k given utterance v is then:

$$P(k|v) = \frac{P(k) \cdot \chi_t(k, v)}{\sum_{k' \in K} P(k') \cdot \chi_t(k', v)} \quad (2)$$

where $P(k)$ is the prior probability of knowledge state k , and $\chi_t(k, v) = 1$ if $v \in t(k)$ and 0 otherwise.

Motivation-based strategies. The set M represents the motivation-based strategies a speaker might follow. Each strategy $m \in M$ is a function $m : C \times K \rightarrow V$ that specifies the utterance a speaker would choose in context c with knowledge state k when driven by a particular motivation. For example, a maximally informative strategy always selects the most precise available utterance given the current knowledge state, while a speaker-convenience strategy always selects the least effortful form regardless of context.

Given a set of strategies, the listener can calculate $P(m|k, c, v)$, the conditional probability that the speaker follows strategy m given knowledge state k , context c , and observed utterance v :

$$P(m|k, c, v) = \frac{P(m) \cdot \chi_m(c, k, v)}{\sum_{m' \in M} P(m') \cdot \chi_m(c, k, v)} \quad (3)$$

where $P(m)$ is the prior probability of strategy m , and $\chi_m(c, k, v) = 1$ if $m(c, k) = v$ and 0 otherwise.

The impact function. The impact function $I(a|k, m; \omega)$ defines how inferences about knowledge state k and motivation strategy m affect the social evaluation on attribute a . It captures the intuition that listeners' assumptions about what a speaker knew and why they spoke as they did are systematically linked to evaluations of social attributes. For example, inferring that a speaker knew a precise value may positively affect perceptions

of competence, while inferring that a speaker chose to omit relevant information may negatively affect perceptions of warmth. Formally:

$$I(a|k, m; \omega) = \omega \cdot \phi_K(k, a) + (1 - \omega) \cdot \phi_M(m, a) \quad (4)$$

where $\phi_K : K \times A \rightarrow \mathbb{T}_3$ and $\phi_M : M \times A \rightarrow \mathbb{T}_3$ are *trivalent functions* that indicate whether and how a knowledge state or motivation-based strategy influences a social attribute.⁵ The weight parameter $\omega \in (0, 1)$ controls the relative contribution of epistemic versus motivational inference: high ω places more weight on what the speaker knew; low ω places more weight on their inferred motivation.

Relation to RSA. It is worth making explicit why SEM is presented as a distinct model rather than a direct extension of RSA, as Frank and Goodman’s (2012) framework could in principle be extended to include social attributes as additional latent variables. There are two reasons for maintaining a distinct formulation. First, RSA’s standard speaker model is utility-based: the speaker is assumed to maximize expected utility, typically a function of informativeness and utterance cost. SEM replaces this with a discrete set of behavioral strategies, each representing a recognizable communicative motivation (being maximally precise, adapting to context, following the path of least effort, etc.). This makes the strategies enumerable, interpretable, and directly groundable in empirical data on attributed motivations—a methodological advantage that is not easily achieved within the utility-based RSA architecture. Second, the impact function I and the trivalent mappings ϕ_K , ϕ_M have no counterpart in RSA: they constitute the formal machinery for linking epistemic and motivational inference to social evaluation, which is the substantive novel contribution of SEM. That said, SEM is compatible with RSA and could be integrated into it as a downstream module: the social evaluations produced by SEM could serve as inputs to a speaker model in RSA, allowing speakers to strategically shape the evaluative inferences their utterances elicit. We return to this integration possibility in Section 5.

⁵A trivalent function maps inputs to one of three values in $\mathbb{T}_3 = \{-1, 0, 1\}$, representing a significant positive effect, no significant effect, or a significant negative effect respectively. These values are fitted empirically from participant data, as detailed in Section 3.

3 A Case Study on Numerical (Im)Precision

We assess the Social Evaluation Model (SEM) by examining the phenomenon of numerical (im)precision, which refers to the level of detail or granularity at which numerical information is conveyed (Krifka, 2009). For example, a speaker who wishes to communicate the time at which an event occurred might use either of the alternatives shown in (1):

- (1) a. “The accident happened at 8:32.” (precise)
 b. “The accident happened around 8:30.” (approximate)

If a speaker has only approximate knowledge of the relevant facts, then if speaking truthfully they can only use the approximate alternative (1b); but a speaker who knows the precise value might express themselves either precisely or approximately, making this a situation of speaker choice.

Numerical (im)precision serves as an ideal testing ground for SEM, highlighting how linguistic choices reflect both contextual adaptation and social interpretation. Previous research has shown that speakers’ preferred degree of precision varies based on contextual demands and communicative relevance (Henst, Carles, and Sperber, 2002; Mühlenbernd and Solt, 2022). Furthermore, different levels of (im)precision convey distinct social meanings (Beltrama, 2018; Beltrama, Solt, and Burnett, 2022), and listeners’ social evaluations are linked to their inferences about a speaker’s knowledge state and communicative motivations (Author, 2025). These findings suggest that (im)precision offers a rich domain for exploring how epistemic and contextual factors jointly shape pragmatic and social inference.

Henst, Carles, and Sperber (2002) provided a foundational account of how context influences numerical precision, finding that speakers adjust their level of precision to match what is communicatively relevant: round numbers were preferred when exact values were pragmatically unimportant, while precise numbers were used when accuracy mattered. Mühlenbernd and Solt (2022) extended this in a production study where participants systematically adapted their expressions to context—more exact expressions when ac-

curacy was pragmatically important, and more approximate ones when fine detail was unnecessary.

Another line of research reveals that numerical (im)precision conveys social meaning. Beltrama (2018) demonstrated that precise non-round expressions can signal positive attributes like intelligence as well as negative ones such as uptightness. Beltrama, Solt, and Burnett (2022) found that speakers using precise forms were rated higher on status-related attributes (e.g., ‘intelligent’, ‘articulate’) but also on ‘pedantic’, whereas approximate speakers received higher ratings on solidarity-related attributes (e.g., ‘likeable’, ‘laid-back’). These evaluative patterns were context-dependent: the status-enhancing effect of precise forms was stronger in formal “for-the-record” contexts than in casual “bonding” contexts. These findings highlight exactly the kind of interplay between pragmatic expectations, social goals, and contextual norms that SEM aims to capture.

We will use SEM to predict the empirical results of two experiments we conducted, reported in full in Author (2025). These experiments investigated how situational context and speaker knowledge state influence the social meaning of precise and imprecise numerical forms. We employed the matched guise methodology (Campbell-Kibler, 2007), where participants evaluate speakers in different versions or ‘guises’ of the same scenario. In what follows, we summarize the experimental design and results relevant to fitting and testing SEM, then present the model predictions.

3.1 Participants, Design, and Methods

Experiment 1. A total of 371 native English speakers, aged 18–64, with U.S. IP addresses took part via Prolific (payment: £1.20). The experiment used six brief, pre-tested scenarios in which one speaker asks a question and another responds with a numerical answer. Two factors were manipulated in a 2×2 design: the required precision level (HPR: high, LPR: low) and the numerical form (PRECISE or APPROX). An example scenario is shown in Figure 3. The study used a fully between-subjects design ($n \approx 90$ per condition), with each participant assigned one of the six scenarios in one of its four versions. Participants completed two tasks relevant to this study:

LPR: Jamie has a new bicycle and is telling a friend about it. The friend is interested and wants to know more. <i>Friend:</i> “How much did the bicycle cost? I’d love to get one like it.”	HPR: Jamie’s new bicycle was stolen. Fortunately it was insured, so Jamie has called the insurance company. <i>Insurance agent:</i> “How much did the bicycle cost? I’ll start the paperwork right away.”
Jamie: “The bicycle cost \$509.55 (PRECISE) / about \$500 (APPROX).”	

Figure 3: Example scenario (LPR: low precision context; HPR: high precision context) with the numerical forms PRECISE and APPROX. This ‘stolen bike’ scenario is one of six pre-tested scenarios used in the experiment.

- **Task 1:** Participants rated the speaker on six social attributes using a 7-point Likert scale: COMPETENT, KNOWLEDGEABLE, WELL-PREPARED, LIKEABLE, HELPFUL, and PEDANTIC.
- **Task 2:** Participants indicated the speaker’s inferred reason for using a precise [imprecise] expression rather than a more approximate [more precise] alternative, via a free-text fill-in-the-blank format.

Experiment 2. A total of 391 native English speakers (same recruitment criteria) participated. The materials were modified versions of the same six scenarios, in which it was established that the speaker had access to a source of precise information and had consulted it before answering. The same 2×2 design and task structure were used. Data from both experiments were combined to yield eight conditions ($2 \text{ forms} \times 2 \text{ contexts} \times 2 \text{ knowledge states}$). The full experimental details and statistical results are reported in Author (2025); here we focus on the findings relevant to fitting and evaluating SEM.

3.2 Experimental Results

All analyses were performed using R (R Core Team, 2021), with linear mixed effects models fitted via *lmerTest* (Kuznetsova, Brockhoff, and Christensen, 2015), using form, context, and their interaction as fixed factors, random intercepts for scenario, and sum coding for predictors. The results of Task 1 for the attributes COMPETENT, LIKEABLE,

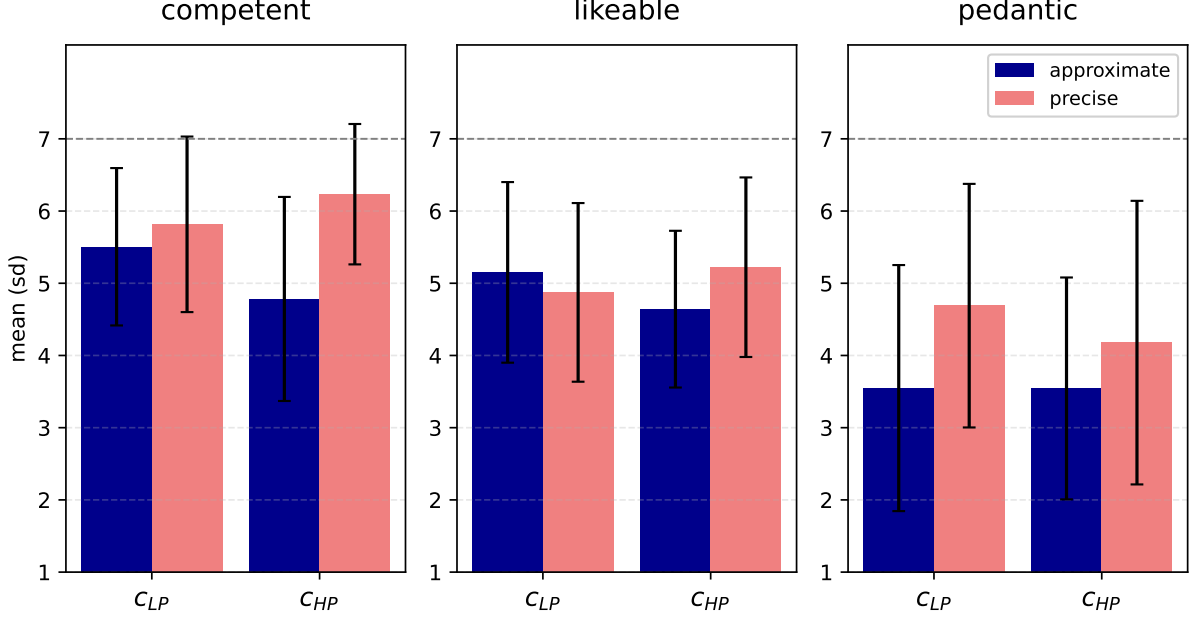


Figure 4: Task 1 rating results (Experiment 1): mean ratings with standard deviations for the attributes COMPETENT, LIKEABLE, and PEDANTIC, by numerical form and context. The dashed line marks the top of the 7-point rating scale.

and PEDANTIC are presented in Figure 4 (mean values and standard deviations).⁶

For COMPETENT and PEDANTIC, a significant main effect of form was found, with PRECISE rated higher ($p < .001$ for both). For COMPETENT and LIKEABLE, a significant interaction of context and form was found: the relative strength of PRECISE versus APPROX was greater in HPR than in LPR ($p < .001$ for both). These results demonstrate that the choice of numerical precision level carries social meaning: precise speakers score higher on competence-related dimensions compared to approximate ones, though they are also perceived as more pedantic. These effects are context-dependent: the positive attributes associated with precise forms are heightened in high-precision situations, while positive associations with speaking approximately are more pronounced in low-precision contexts.

Figure 5 displays the results of Task 2, showing frequencies of the 10 most common motivation classes, color-coded by form (blue: approximate, red: precise). Two coders independently coded the free-text responses; a brief description of each motivation class

⁶The attributes KNOWLEDGEABLE, WELL-PREPARED, and HELPFUL showed closely aligned statistics with COMPETENT across all tasks (see Author, 2025, for full results). We therefore exclude them from individual treatment in the SEM evaluation, treating them as encompassed by COMPETENT.

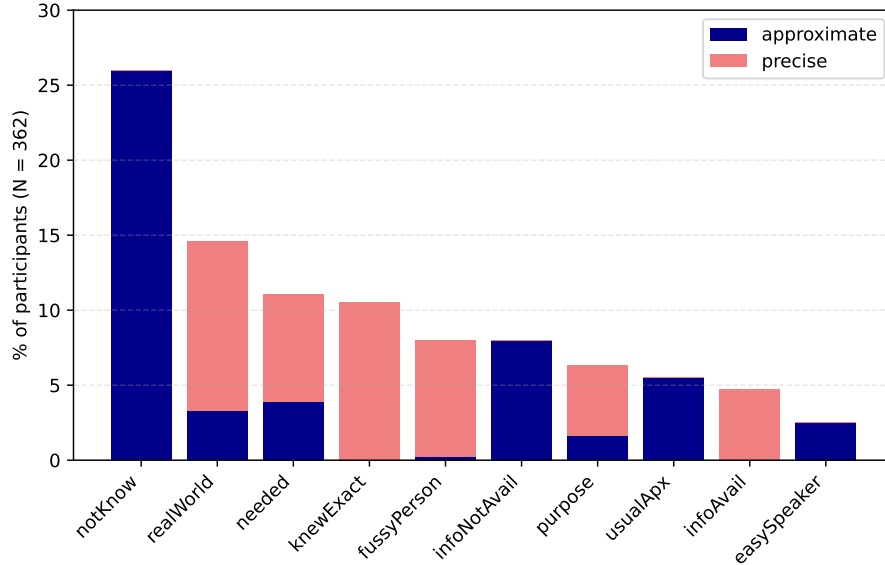


Figure 5: Relative frequencies (as a percentage of all participants) of the 10 most frequently mentioned motivation classes, color-coded by form: approximate (blue), precise (red).

is provided in Figure 7(a).

To explore whether inferred speaker motivations influence social meaning—a core assumption of SEM—we compared, for each Task 2 motivation class, the Task 1 ratings of participants who did versus did not mention that class, using independent-samples t -tests (Experiment 1). Figure 7(b) displays the p -values for all combinations (3 social attributes $\times$ 10 motivation classes), with significant effects ($p < .01$) in bold font and direction indicated by color (green: positive, red: negative). A total of 7 significant effects were found across 30 combinations, supporting the central SEM assumption that social meaning is influenced by assumed speaker motivations.

3.3 Fitting the Model

To fit SEM to this case study, we construct a scenario model that matches the experimental design. We define two situational contexts: c_{LP} (low-precision needs) and c_{HP} (high-precision needs), and two utterance types: v_{prc} (precise) and v_{apx} (explicitly approximate). Figure 6 illustrates the model using an analogous scenario in which a speaker reports the time of an event either precisely (“at 8:32”) or approximately (“around 8:30”), to either a friend at a dinner party or a police officer.

We consider two relevant knowledge states: k_p , the speaker has precise knowledge (e.g., knows the event happened at 8:32), and $k_{\neg p}$, the speaker has only approximate knowledge (e.g., knows it happened around $8:30 \pm 5$ min). The truthfulness function t is defined as $t(k_p) = \{v_{apx}, v_{prc}\}$ and $t(k_{\neg p}) = \{v_{apx}\}$: a speaker with only approximate knowledge cannot truthfully use a precise utterance. This is shown in Figure 6(a).

We consider three motivation-based strategies, shown in Figure 6(b):

- m_{max} : the speaker is always maximally precise, as long as their knowledge state permits it (always precise if k_p , approximate if $k_{\neg p}$).
- m_{sit} : the speaker matches their precision to the situational demands (precise in c_{HP} , approximate in c_{LP} , as long as knowledge permits).
- m_{min} : the speaker is always minimally precise regardless of context or knowledge.

Figure 6(c) illustrates the listener’s inferences after hearing a precise utterance: the listener concludes the speaker has precise knowledge (k_p), and in a high-precision context assigns positive probability to both m_{max} and m_{sit} , while in a low-precision context infers the speaker follows only m_{max} .

To fit the impact function I , we examine the Task 2 results. The full fitting procedure is outlined in Figure 7. First, we mapped the motivation classes from Task 2 (Figure 7(a)) to model parameters. For knowledge states: KNEWEXACT and INFOAVAIL map to k_p ; NOTKNOW and INFONOTAVAIL map to $k_{\neg p}$. For motivation-based strategies: FUSSYPERSON (inherently precise) maps to m_{max} ; EASYSPKAKER and USUALAPX (consistent approximation for non-contextual reasons) map to m_{min} ; NEEDED, REALWORLD, and PURPOSE (contextually-sensitive motivations) map to m_{sit} . All mappings are listed in Figure 7(c).

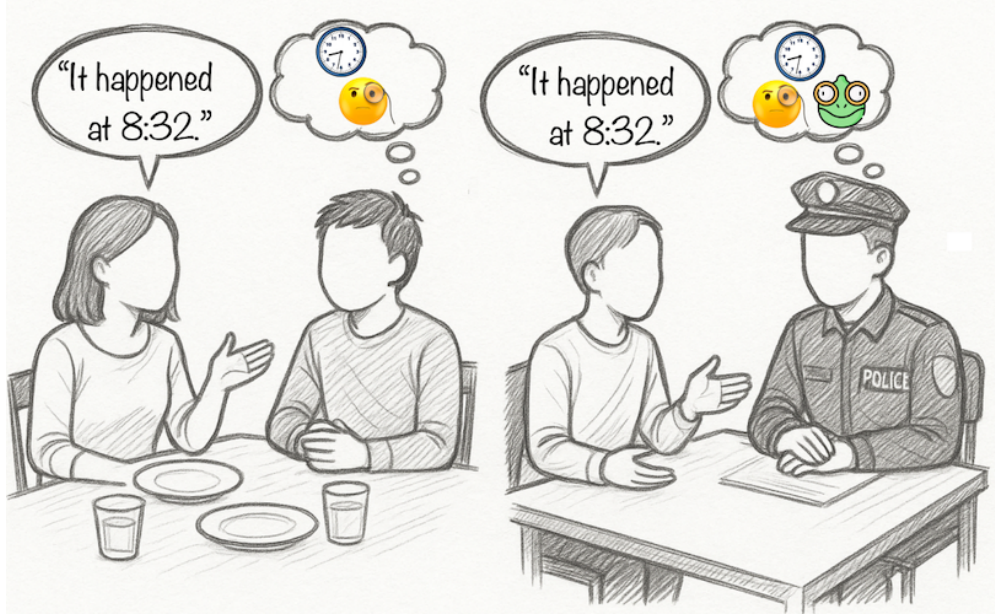
The trivalent functions ϕ_K and ϕ_M are then derived from these comparisons (Figure 7(b)): for each motivation class and attribute we assign +1 for a significant positive difference, −1 for a significant negative difference, and 0 otherwise (threshold $p < .01$), and replace motivation class labels with the corresponding model parameters per the

knowledge states $k \in K$	respective truthful utterances $v \in t(k)$
k_p 	v_{apx} “It happened around 8:30.” v_{prc} “It happened at 8:32.”
$k_{\neg p}$ 	v_{apx} “It happened around 8:30.”

(a) speaker’s knowledge states and respective truthful utterances

	c_{HP} (police)		c_{LP} (dinner)	
	k_p	$k_{\neg p}$	k_p	$k_{\neg p}$
m_{max} 	v_{prc}	v_{apx}	v_{prc}	v_{apx}
m_{sit} 	v_{prc}	v_{apx}	v_{apx}	v_{apx}
m_{min} 	v_{apx}	v_{apx}	v_{apx}	v_{apx}

(b) motivation-based strategies $m : C \times K \rightarrow V$



(c) model predictions for which knowledge states and motivation-based strategies a listener considers as probable after a speaker utters v_{prc} (left: dinner context; right: police context)

Figure 6: Exemplary scenario: a speaker reports the time of an event (8:32) to a friend at a dinner party (low-precision context) or a police officer (high-precision context). (a) Speaker knowledge states and respective truthful utterances. (b) The three motivation-based strategies: m_{max} (always maximally precise), m_{sit} (situation-adapted), m_{min} (always minimally precise). (c) Listener inferences after hearing a precise utterance, by context.

mappings in Figure 7(c). Where several motivation classes map to the same parameter, the cell is set by majority vote. The resulting values are shown in Figure 7(d).⁷

⁷One cell— $\phi_M(m_{sit})$ on COMPETENT—lies at the significance boundary: the underlying NEEDED–COMPETENT difference is significant under a two-sample t -test ($p = .010$) but falls just short under a mixed-effects model with random intercepts for scenario ($p = .012$). We retain the t -test value; the model’s four qualitative predictions hold under either choice.

Class	Description
KNEWEXACT	<i>S</i> knew/was confident about exact/precise information
NOTKNOW	<i>S</i> did not know, not sure/certain, not remember, approx. knowledge
FUSSYPERSON	<i>S</i> is anal/predantic/detail oriented
EASYSPEAKER	using <i>apx</i> was easier/faster for <i>S</i>
USUALAPX	<i>S</i> usually uses <i>apx</i>
NEEDED	utterance is needed, required, called for, important, necessary, or sufficient
REALWORLD	real-world consequences of communicated information
PURPOSE	<i>S</i> is purposefully precise/approximate
INFONOTAVAIL	precise information was not available
INFOAVAIL	precise information was available

(a) The 10 most frequent motivation classes



Parameter	Motivation class(es)
k_p	KNEWEXACT, INFOAVAIL
$k_{\neg p}$	NOTKNOW, INFONOTAVAIL
m_{max}	FUSSYPERSON
m_{min}	EASYSPEAKER, USUALAPX
m_{sit}	NEEDED, REALWORLD, PURPOSE

(c) Mappings from motivations to model parameters



	LIKEABLE	COMPETENT	PEDANTIC
KNEWEXACT	.480	<.001	.028
NOTKNOW	.189	<.001	<.001
FUSSYPERSON	.016	.033	<.001
EASYSPEAKER	.062	.003	.199
USUALAPX	.933	.129	.106
NEEDED	.331	.010	.131
REALWORLD	.365	.098	.857
PURPOSE	<.001	.034	.252
INFONOTAVAIL	.507	.851	.096
INFOAVAIL	.185	.116	.015

(b) p values for effects of motivation on feature evaluations



	a_{like}	a_{comp}	a_{ped}
m_{max}	0	0	1
m_{sit}	1	1	0
m_{min}	0	-1	0

Fitted values for $\phi_M(m, a)$

	a_{like}	a_{comp}	a_{ped}
k_p	0	1	0
$k_{\neg p}$	0	-1	-1

Fitted values for $\phi_K(k, a)$

(d) Resulting trivalent functions

Figure 7: (a) The 10 most frequent motivation classes with descriptions. (b) p -values for associations between motivation classes and attribute ratings (COMPETENT, LIKEABLE, PEDANTIC); significant effects ($p < .01$) in bold, green (positive) or red (negative). (c) Mappings from motivation classes to model parameters. (d) Resulting trivalent functions ϕ_K and ϕ_M .

3.4 Model Predictions

The fitted SEM has the following free parameters: weight $\omega \in (0, 1)$ in the impact function (Eq. 4), and the prior probabilities $P(k)$ and $P(m)$ used to compute the posteriors in Equations 2 and 3.

For an initial model check, we test a *balanced model*: $\omega = 0.5$ (equal weight to epistemic and motivational inference) and flat prior probabilities.⁸ We then evaluate the model across a wide range of parameter combinations to assess robustness.

The model check tests whether SEM reproduces the four significant effects from Experiment 1 (all $p < .001$):

1. PRECISE is rated higher than APPROX on COMPETENT across all contexts.

⁸Specifically, $P(m_{max}) = P(m_{sit}) = P(m_{min}) = \frac{1}{3}$ and $P(k_p) = P(k_{\neg p}) = \frac{1}{2}$.

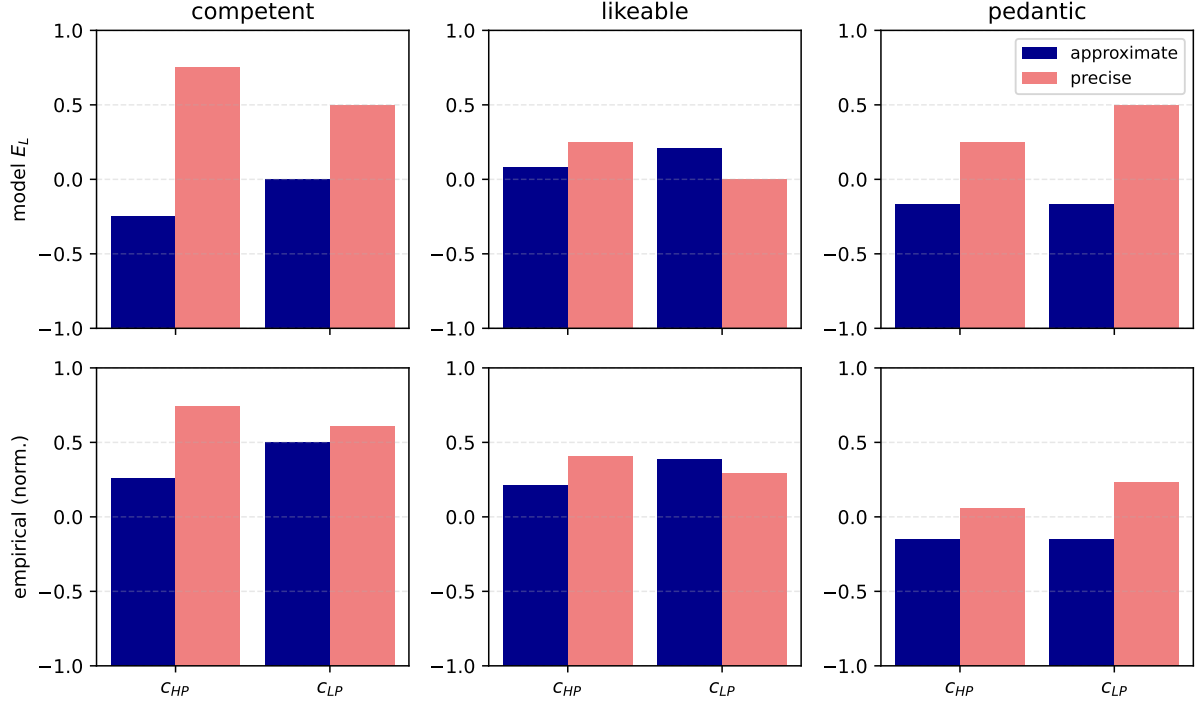


Figure 8: Predictions of the balanced model for the attributes COMPETENT, LIKEABLE, and PEDANTIC (top row, evaluation scores $E_L \in [-1, 1]$), shown alongside the empirical results of Experiment 1 (bottom row, mean ratings linearly rescaled to $[-1, 1]$ via $(x - 4)/3$). Bars rise from the scale floor (-1 , the lowest evaluation); 0 denotes a median—not neutral—evaluation.

2. PRECISE is rated higher than APPROX on PEDANTIC across all contexts.
3. The positive difference between PRECISE and APPROX on COMPETENT is greater in HPR than in LPR.
4. The relative strength of APPROX versus PRECISE on LIKEABLE is greater in LPR than in HPR.

SEM is considered qualitatively correct if it reproduces the direction of these effects:

1. $E_L(a_{comp}|v_{prc}, c_{HP}) + E_L(a_{comp}|v_{prc}, c_{LP}) > E_L(a_{comp}|v_{apx}, c_{HP}) + E_L(a_{comp}|v_{apx}, c_{LP})$
2. $E_L(a_{ped}|v_{prc}, c_{HP}) + E_L(a_{ped}|v_{prc}, c_{LP}) > E_L(a_{ped}|v_{apx}, c_{HP}) + E_L(a_{ped}|v_{apx}, c_{LP})$
3. $E_L(a_{comp}|v_{prc}, c_{HP}) - E_L(a_{comp}|v_{apx}, c_{HP}) > E_L(a_{comp}|v_{prc}, c_{LP}) - E_L(a_{comp}|v_{apx}, c_{LP})$
4. $E_L(a_{like}|v_{prc}, c_{HP}) - E_L(a_{like}|v_{apx}, c_{HP}) > E_L(a_{like}|v_{prc}, c_{LP}) - E_L(a_{like}|v_{apx}, c_{LP})$

Figure 8 presents the balanced model’s predictions for a_{comp} , a_{like} , and a_{ped} across utterances and contexts. The balanced model correctly predicted all four effects. The qualitative alignment between model predictions and empirical results is visible in Figure 8; the attribute-internal patterns are effectively reproduced.

To assess robustness, we tested the balanced model against an exhaustive grid of free parameter combinations: every combination of ω (from 0.05 to 0.95) and the prior probabilities $P(k_p)$, $P(k_{\neg p})$, $P(m_{max})$, $P(m_{sit})$, and $P(m_{min})$ (from 0 to 1), each in increments of 0.02, yielding 2,593,080 valid combinations. SEM correctly predicted all four effects for 99.87% of combinations; for the remaining 0.13% it correctly predicted three out of four.⁹ This demonstrates the robustness of the qualitative predictions across the free parameter space.

Section 4 presents a quantitative evaluation of these predictions, comparing the full model against nested ablations to assess which inferential components are necessary to reproduce the empirical patterns.

4 Quantitative Evaluation

The preceding sections established that SEM qualitatively reproduces the four significant empirical effects from Experiment 1 across almost all free parameter settings (99.87% of 2,593,080 combinations). This section asks a deeper question: *how well* does the model reproduce those effects, and does its formal inference machinery add explanatory value over simpler alternatives?

We address this with two analyses. First, we evaluate the balanced model’s quantitative fit against the empirical means using an established metric suite. Second, we compare the full model against three nested ablations—each of which removes one inferential component—evaluated across a sample of parameter combinations rather than a single setting, so that differences reflect structural properties of the model variants rather than parameter choices.

⁹In particular, the model struggled to predict Effect 1 correctly when ω was very low, $P(k_p)$ was very low, and $P(m_{sit})$ was high—a combination of parameters that is difficult to motivate on independent grounds.

4.1 Metric Framework

Quantitative model evaluation follows the framework proposed by **muehlenbergLLM2026** for assessing probabilistic pragmatic models against human data. We compare model scores $E_L(a|v, c) \in [-1, 1]$ against empirical human means, which are linearly rescaled from the 7-point Likert scale to $[-1, 1]$ via $(x - 4)/3$, making model and human values directly comparable.

Six effects are assessed for each of the three SEM-modeled attributes (a_{comp} , a_{like} , a_{ped}): a main effect of form (aggregated across contexts) and a form $\times$ context interaction. Of these six effects, four are statistically significant in the human data (both main effects: a_{comp} and a_{ped} ; both interactions: a_{comp} and a_{like}); two are non-significant (main effect of a_{like} ; interaction of a_{ped}).

We report the following metrics:

- **DAS** (Directional Agreement Score): proportion of *significant* main effects on which model and human agree in sign. Range $[0, 1]$; 1 = perfect directional agreement on all significant main effects.
- **ISS** (Interaction Sensitivity Score): same calculation applied to significant form $\times$ context interactions. Range $[0, 1]$.
- **ESR** (Effect Size Ratio): for each *significant* effect, $|\Delta_M|/|\Delta_H|$ where Δ denotes the model or human contrast value. $ESR = 1$ means perfect magnitude calibration; $ESR > 1$ means the model overshoots. Computed only for significant effects to avoid division by near-zero human contrasts.
- **CDS** (Calibration Deviation Score): mean $|ESR - 1|$ across significant effects. Lower is better; 0 = perfect magnitude calibration.
- **RMSE**: root mean squared error computed cell-by-cell over all 12 condition means (3 attributes $\times$ 2 forms $\times$ 2 contexts), after rescaling human means to $[-1, 1]$. Lower is better.

All metrics are restricted to the four statistically significant effects, with the exception of RMSE, which is computed over all 12 conditions to provide a global picture of fit across the full data surface. Testing directional agreement or magnitude calibration on non-significant effects would be analogous to computing ESR on a near-zero human contrast: the sign and magnitude of such effects are noise, not signal.

4.2 Balanced Model Fit

Table 1 reports fit metrics for the balanced model ($\omega = 0.5$, flat priors).

Table 1: Quantitative fit of the balanced model ($\omega = 0.5$, flat priors) against empirical means from Experiment 1 ($N = 362$). DAS, ISS, ESR, and CDS are computed over the four statistically significant effects only; RMSE over all 12 conditions.

Metric	Value
RMSE	0.26
DAS	1.00
ISS	1.00
ESR main a_{comp} / a_{ped}	2.55 / 1.83
ESR interaction a_{comp} / a_{like}	1.32 / 1.31
CDS (main)	1.19
CDS (interaction)	0.31
CDS (overall)	0.75

Two findings stand out. First, the balanced model achieves perfect directional agreement on all significant effects: $DAS = 1.00$ and $ISS = 1.00$ mean that it correctly reproduces the sign of both significant main effects (a_{comp} , a_{ped}) and both significant interactions (a_{comp} , a_{like}). This is the model’s strongest result, because the interactions reflect the context-sensitive pragmatic mechanism at the heart of SEM. Second, the model overshoots effect magnitudes on main effects: ESR values of approximately $2\times$ indicate that SEM predicts roughly twice the human contrast for both a_{comp} and a_{ped} . This drives CDS (main) to 1.19. CDS for interactions (0.31) is substantially better, reflecting more accurate magnitude calibration where the model’s context-sensitive reasoning does the most work.

The magnitude overshoot on main effects is a property of the balanced parameter setting rather than a structural limitation of the model. A grid search over the free

parameter space identifies a setting ($\omega = 0.7$, $P(k_p) = 0.9$, $P(m_{max}) = 0.5$, $P(m_{sit}) = 0.3$, $P(m_{min}) = 0.2$) that achieves near-perfect calibration (CDS = 0.056, ESR values of 1.03, 1.01, 1.15, 0.96; RMSE = 0.165), while preserving ISS = 1.00. This confirms that the overshoot under balanced parameters reflects the uninformative prior choice, not an architectural flaw.¹⁰

4.3 Model Variant Comparison

A central reviewer concern is whether the full SEM’s inference machinery adds explanatory value beyond simpler alternatives. We address this by comparing the full model against three nested ablations, each of which removes one inferential component while holding everything else equal:

- **SEM-full**: the complete model, all parameters free.
- **SEM-context-blind**: $P(m_{sit}) = 0$, removing the context-sensitive motivation strategy. This operationalizes a “violation-of-expectation” account in which the model can infer knowledge states and non-contextual motivations, but has no mechanism for context to modulate evaluations positively.
- **SEM-knowledge-only**: $\omega = 1$, so only epistemic inference contributes to evaluation. Motivation inference is disabled entirely.
- **SEM-motivation-only**: $\omega = 0$, so only motivational inference contributes. Knowledge-state inference is disabled entirely.

To make the comparison robust to parameter choice, each variant is evaluated across 100 sampled parameter combinations rather than a single setting. Free parameters are drawn from uninformative distributions ($\omega \sim \text{Beta}(2, 2)$ clipped to $(0.05, 0.95)$; $P(k) \sim \text{Dirichlet}(1, 1)$; $P(m) \sim \text{Dirichlet}(1, 1, 1)$; seed fixed for reproducibility), and metrics are computed for each draw. We report mean $\pm$ standard deviation across draws. A difference

¹⁰This is an in-sample, post-hoc result: the optimized parameters are tuned to the same 12 data cells from which the impact functions were derived, so it represents an upper bound on fit rather than a validated generalization. The balanced model is the primary evaluation throughout; the grid search result is reported as a calibration bound.

that is consistent across the full sample reflects a structural property of the model variant, not a parameter artifact.

Table 2 reports the results.

Table 2: Quantitative comparison of four SEM variants across 100 sampled parameter combinations (mean $\pm$ SD). DAS, ISS, ESR, and CDS are computed over the four significant effects only; RMSE over all 12 conditions. Lower CDS and RMSE are better; higher ISS and DAS are better.

Metric	SEM-full	Context-blind	Know-only	Motiv-only
DAS	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.96 $\pm$ 0.13
ISS	1.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.04 $\pm$ 0.14	1.00 $\pm$ 0.00
RMSE	0.30 $\pm$ 0.10	0.42 $\pm$ 0.12	0.39 $\pm$ 0.14	0.51 $\pm$ 0.09
CDS (main)	1.06 $\pm$ 0.71	1.46 $\pm$ 0.72	1.65 $\pm$ 1.25	1.07 $\pm$ 0.56
CDS (inter)	0.82 $\pm$ 0.62	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.79 $\pm$ 1.44
CDS (overall)	0.94 $\pm$ 0.43	1.23 $\pm$ 0.36	1.32 $\pm$ 0.62	1.43 $\pm$ 0.67

The key finding is ISS: it collapses to ≈ 0 for the two variants that remove m_{sit} (context-blind: 0.00; knowledge-only: 0.04) and stays at 1.00 for the two that retain it (full and motivation-only). Since the two significant interaction effects are precisely the form $\times$ context effects captured by m_{sit} , this means context-blind and knowledge-only cannot reproduce two of the four significant empirical effects at all, regardless of parameter setting. The context-sensitive motivation mechanism is structurally necessary, not a decorative feature of the model.

Each ablation fails on a distinct dimension, and none matches the full model on both structure and calibration. The context-blind variant loses the interactions entirely (ISS = 0) and also degrades on RMSE (0.30 $\rightarrow$ 0.42) and overall CDS (0.94 $\rightarrow$ 1.23). The motivation-only variant keeps the interaction directions (ISS = 1.00) but is otherwise the weakest: it massively overshoots interaction magnitudes (mean ESR $\approx$ 2.7, CDS interaction = 1.79) and has the highest RMSE (0.51), because without knowledge inference the competence ordering breaks down. The knowledge-only variant achieves the second-lowest RMSE (0.39), reflecting that the CS1 overall ordering is largely epistemic—precise speakers are inferred to have precise knowledge, which is the dominant signal in the data. But its ISS of 0.04 means it fails entirely on the interactions. Without motivational in-

ference, the model cannot represent context as modulating the epistemic ordering, and so it misses two of the four significant empirical effects regardless of parameter setting. A model that reproduces the main trend while missing the context-sensitivity has not captured the phenomenon.

Taken together, the full SEM is the only variant that reproduces the interaction structure ($ISS = 1.00$) while remaining competitive on global fit (lowest RMSE, lowest overall CDS). Every simpler alternative either matches or exceeds it on one metric while failing conspicuously on another. This pattern provides direct evidence that the model’s staged inference—from epistemic state, through contextual motivation, to social attribute—is necessary rather than redundant.

The CDS results also bear on the question of magnitude calibration raised above. The interaction CDS difference between the full model (0.82) and the ablations that lose m_{sit} (both pinned at 1.00, since their predicted interaction contrasts are zero) quantifies the specific contribution of contextual motivation inference to calibration. The full model’s main-effect CDS (1.06) is essentially tied with the motivation-only variant (1.07), confirming that main-effect overshoot is a shared property of the trivalent impact scale and not specific to any particular model configuration.

Threshold robustness. The ϕ functions are derived at $p < .01$. Re-running the full comparison with ϕ derived at $p < .05$ (which adds three non-zero cells) worsens the full model on every calibration metric (CDS $0.94 \rightarrow 1.45$, RMSE $0.30 \rightarrow 0.51$) while leaving ISS unchanged. Tightening to $p < .001$ slightly improves CDS ($0.94 \rightarrow 0.89$) but degrades interaction coverage ($ISS\ 1.00 \rightarrow 0.57$) because the borderline $m_{sit}-a_{comp}$ cell is dropped. The $p < .01$ threshold is the only one that maintains full interaction coverage while achieving competitive calibration.¹¹

¹¹These threshold comparisons were conducted by re-deriving ϕ_K and ϕ_M at each threshold from the same Experiment 1 data and re-running the variant comparison. Scripts are available in the repository.

5 Discussion and Conclusion

The Social Evaluation Model (SEM) is a probabilistic framework that explains how listeners form social impressions of speakers based on their pragmatic behavior, integrating cognitive, epistemic, and social reasoning within a unified formal architecture. By extending the inferential chain of probabilistic pragmatics, SEM shows how linguistic choices—such as the level of numerical precision—give rise to evaluations of traits like competence, warmth, and pedantry through staged Bayesian inference. The model accurately reproduces the key empirical patterns from a matched-guise study on numerical (im)precision: it predicts how varying levels of precision affect competence, likeability, and pedantry ratings as a function of contextual demands, and the quantitative evaluation demonstrates that its staged inference architecture—from epistemic state, through contextual motivation, to social attribute—is necessary to reproduce the observed pattern. Three nested ablations each fail on a structurally distinct dimension, with the context-blind variant providing the clearest evidence: removing the context-sensitive motivation strategy collapses interaction sensitivity to zero, directly demonstrating that the m_{sit} mechanism is not decorative.

Relation to the RSA framework. The relationship between SEM and the Rational Speech Act framework (Frank and Goodman, 2012; Goodman and Frank, 2016) deserves careful discussion. A recurring question is whether a new model is needed, or whether the existing RSA architecture could be extended to cover social evaluation. We believe there are principled reasons for maintaining SEM as a distinct model, alongside its natural compatibility with RSA.

RSA is a framework for modeling how communicative meaning is derived through recursive reasoning about speaker and listener mental states. In standard RSA, the speaker selects utterances to maximize a utility function, typically a function of informational value and cost; the listener uses Bayes’ rule to invert this speaker model and recover the intended meaning. Social goals can be incorporated into the utility function—as in Yoon et al.’s (2020) model of polite speech, which treats politeness as a competing goal along-

side informativeness. This shows that RSA can be extended to encompass some aspects of social reasoning, and we take this as an important point of contact.

SEM differs, however, in its architecture and target of inference. First, SEM models social evaluation from the *listener's* perspective, whereas RSA typically models the speaker's production choices. In SEM, the listener's inferential process yields a social evaluation as its output; the model does not make claims about how speakers optimize their utterances (though see below for how SEM and RSA can be integrated). Second, SEM replaces the continuous utility function of RSA with a discrete set of behavioral strategies, each representing a recognizable communicative motivation. This design choice makes the strategies directly groundable in empirical data—specifically, in the motivation classes that participants attribute to speakers in free-text tasks—and makes the model's parameters interpretable and testable. A utility-based extension of RSA could also include social attributes as latent variables, but the mapping from utilities to the behavioral diversity captured by SEM's motivation strategies would need to be explicitly specified. Third, the impact function I and the fitted trivalent functions ϕ_K and ϕ_M have no counterpart in standard RSA. They constitute the formal machinery for linking pragmatic inference to social evaluation, and their empirical grounding is a methodological contribution distinct from RSA extensions.

Taken together, SEM is best understood not as an alternative to RSA but as a complementary module that extends the inferential chain one step further: from what the speaker meant, to what kind of communicator they are.

Relation to the Social Meaning Game. SEM also relates to the Social Meaning Game (SMG) framework (Burnett, 2017; Burnett, 2019; Burnett, 2023). SMG models how linguistic variation acquires indexical social meaning through game-theoretic reasoning: speakers and listeners reason about the mapping between linguistic forms and social categories, and social meaning arises from learned indexical links between form and identity.

SEM and SMG are complementary in scope. SMG is designed for semantically equivalent variants whose social meaning is form-based (e.g., sociophonetic variation such as

ING); SEM is designed for semantically non-equivalent alternatives where social meaning arises from pragmatic inference about the speaker’s epistemic state and motivation. SMG explains how forms come to index social categories; SEM models how differences in semantic content or pragmatic choice lead listeners to attribute specific character traits. Correspondingly, SMG is better suited to explaining the acquisition and conventionalization of social meanings over time, while SEM is better suited to explaining how novel or context-sensitive pragmatic choices give rise to social evaluations in the moment of interpretation.

Contributions and limitations. The case study and quantitative evaluation, taken together, demonstrate SEM’s potential as a principled framework for modeling the social consequences of pragmatic behavior. The model’s core contribution is its formalization of social evaluation as an explicit target of probabilistic inference, grounded in empirical data on attributed motivations. The staged inference architecture—from utterance to knowledge state, to motivation, to social attribute—is transparent, testable, and quantitatively evaluable. The variant comparison provides direct evidence that each inferential component contributes independently to the model’s predictions.

Several limitations of the current formulation are worth noting. The trivalent functions ϕ_K and ϕ_M make a simplifying assumption: that a given knowledge state or motivation-based strategy has a fixed, consistent impact on a social attribute regardless of context. In reality, the underlying motivations corresponding to a single behavioral strategy may differ in their social implications. A more refined model would include an additional inference step recovering the probability of each specific motivation given a strategy; we leave this generalization to future work. The current model has been evaluated on a single empirical case study (numerical imprecision); the framework’s applicability to other pragmatic domains—for example, politeness phenomena or relevance violations—is a natural direction for future empirical work.

Integration with RSA and future directions. Looking forward, SEM can be naturally integrated into higher-order models of pragmatic reasoning such as RSA. In such

an integrated model, the outcomes of SEM—listeners’ judgments of competence, warmth, or sincerity—become part of the speaker’s production process: speakers could anticipate the evaluative inferences their utterances are likely to elicit, and strategically choose expressions that present them favorably. This connects pragmatic reasoning to social signaling and impression management, framing communication as recursive social cognition in which agents reason continuously about both the informational and social consequences of their linguistic choices. Developing this integration formally, and testing its predictions empirically, is a central direction for future research.

Data and code availability. The experimental data and R analysis scripts are available at [OSF LINK TO ADD]. A Python implementation of the SEM and its application to both case studies is available at [GITHUB LINK TO ADD].

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